

C++20 MESH REDUCTION · GAME-ENGINE IMPORT PIPELINES

Fewer triangles. Measured transitions.

Round 4 starts at 512 pixels and follows real meshes down to 16. Better candidate search reduces late triangles; exact repeats share one runtime level.

Geometry, with an error budget.

Propose a simpler mesh. Check it against the previous level and the original. Keep the smallest valid chain found.

FROZEN MESH CORPUS 120 30 rocks · 30 organics 40 manufactured · 20 complex scans	512 → 16 px / COVERAGE PILOT 80.18 SCORE +4.31 points · same eight assets and visual gates	FROZEN VALIDATION / 20 MESHES 86.48 SCORE +6.18 points · 20/20 completed · 0 failures	INTEGRATION PATHS 3 chain modes Direct · progressive · hybrid Rebuild vertices or reuse source buffers
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01 / ACTUAL OUTPUT GEOMETRY

From source to 16 pixels.

Four examples from the recorded hybrid coverage run. Switch to Runtime levels to hide exact repeats.

DISPLAY	ClayWireframeSilhouette	LEVELS	Audit slotsRuntime levels	SCALE	EnlargedTarget px	Reset views ↗
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Enlarged for inspection. Same camera and source scale across each row. Drag a mesh to rotate its chain; click a tile to inspect it.

01 / ROCK

Moon rock

Rock surface detail gives way to the permitted distant silhouette. · 8 runtime levels / 8 audit slots

L0D 0512.0 px

L0D 1312.1 px

L0D 2198.2 px

L0D 3115.9 px

L0D 478.7 px

L0D 543.1 px

L0D 626.3 px

L0D 716.0 px

3,304 tris
UNCHANGED SOURCE

2,974 tris
✓ Δ 1.41 / 2.00

2,676 tris
✓ Δ 1.96 / 2.17

268 tris
✓ Δ 2.31 / 2.33

134 tris
✓ Δ 2.47 / 2.50

68 tris
✓ Δ 2.16 / 2.67

34 tris
✓ Δ 2.46 / 2.83

18 tris
✓ Δ 2.16 / 3.00

−99.5%

at LOD 7

3,304 → 18 tris

02 / ORGANIC

Quiver tree trunk

The extreme tail passes pixel-distance gates but loses filled area. · 8 runtime levels / 8 audit slots

L0D 0512.0 px

L0D 1312.1 px

L0D 2198.2 px

L0D 3115.9 px

L0D 478.7 px

L0D 543.1 px

L0D 626.3 px

L0D 716.0 px

17,978 tris
UNCHANGED SOURCE

1,066 tris
✓ Δ 1.98 / 2.00

534 tris
✓ Δ 2.05 / 2.17

268 tris
✓ Δ 2.29 / 2.33

134 tris
✓ Δ 2.12 / 2.50

68 tris
✓ Δ 2.51 / 2.67

6 tris
✓ Δ 2.72 / 2.83

2 tris
✓ Δ 1.46 / 3.00

−99.99%

at LOD 7

17,978 → 2 tris

03 / MANUFACTURED

Metal stool

This asset still needs better intermediate proposals. · 6 runtime levels / 8 audit slots

L0D 0512.0 px

L0D 1312.1 px

L0D 2198.2 px

L0D 3115.9 px

L0D 478.7 px

L0D 543.1 px

L0D 626.3 px

L0D 716.0 px

6,532 tris
UNCHANGED SOURCE

1,981 tris
✓ Δ 1.71 / 2.00

990 tris
✓ Δ 1.96 / 2.17

495 tris
✓ Δ 1.77 / 2.33

247 tris
✓ Δ 2.47 / 2.50

247 tris
✓ Δ 0.00 / 2.67

124 tris
✓ Δ 2.16 / 2.83

124 tris
✓ Δ 0.00 / 3.00

−98.1%

at LOD 7

6,532 → 124 tris

04 / COMPLEX SCAN

Bell X-1

Repeated audit slots can share one runtime mesh. · 5 runtime levels / 8 audit slots

L0D 0512.0 px

L0D 1312.1 px

L0D 2198.2 px

L0D 3115.9 px

L0D 478.7 px

L0D 543.1 px

L0D 626.3 px

L0D 716.0 px

99,586 tris
UNCHANGED SOURCE

10,180 tris
✓ Δ 1.46 / 2.00

10,180 tris
✓ Δ 0.00 / 2.17

5,090 tris
✓ Δ 2.05 / 2.33

2,546 tris
✓ Δ 1.71 / 2.50

2,546 tris
✓ Δ 0.00 / 2.67

1,273 tris
✓ Δ 2.83 / 2.83

1,273 tris
✓ Δ 0.00 / 3.00

−98.7%

at LOD 7

99,586 → 1,273 tris

Reading the tiles: Δ is the audited upper error against the previous LOD, followed by its pixel limit. Each reduced mesh also passed its direct-to-source limit. LOD 0 is unchanged.

Neutral clay uses geometric face normals, without textures. Presentation angles are not audit cameras. A low coverage error does not constrain shading.

02 / COMPARABLE COVERAGE SCORES

Useful final LODs need better search.

Preventing new topology locks helps coarse reductions. Avoiding duplicate hybrid inputs and using the full existing proposal budget improves the complete chain score.

Previous implementation		75.86
Relax topology locks		75.68
Prevent new locks		75.75
Guard + better search		80.18
Correct input ownership		80.18

075100

Same eight development assets, 512 → 16 px, hybrid rebuild, source cap 8 px, eight proposals per level, unchanged cameras and gates. SCORE = 100 × (1 − category-balanced mean retained triangle ratio). Exact duplicate compaction does not change SCORE.

03 / WHAT THE EXPERIMENTS SAY

Smaller tails, with measured limits.

−48.9%

The last levels use fewer triangles.

Category-balanced final retained triangles fall from 3.14% to 1.61%. Last-three-level retained ratio falls 50.9%. Gates and SCORE are unchanged.

FINAL RETAINED RATIO

53/64

Exact repeats share one mesh.

The C/C++ API and glTF export expose compact runtime levels. Every scheduled source and transition audit stays in the manifest and in the score.

RUNTIME / AUDIT SLOTS

2 vs 34

A loose distance budget can lose filled area.

The 2-triangle tail passes the denser 706-camera recheck but loses much of the thin trunk's coverage. Its smaller quad cost includes that coverage loss. Shading is outside this coverage-only run.

TREE / CAP 8 vs 2

THE VISUAL CONTRACT 512 → 16 px · 8 audit slots 2 → 3 px transitions; source cap 8 px. The cumulative policy reaches 6.89 px at the final level. Runtime meshes share exact duplicate slots.	THE AUDIT BEHIND THESE EXAMPLES 12 orthographic + 4 perspective views Coverage is checked at 4× sampling with refinement to 8×. Search uses a separate camera set. These are finite sampled views, not a universal error guarantee.	WHAT REMAINS OUTSIDE THIS BOARD Full-quality corpus audit The default 642 + 64 camera preset has not been run across the full corpus. Texture images, opacity, skinning and morph targets are outside this version.
BLITZREMESHER / ROUND 4 C++20 · exact runtime compaction · quad occupancy proxies Round 4 report · Validation · Provenance		Actual meshes. C00 source assets. Moon rock · Quiver tree trunk · Metal stool · Bell X-1 Four illustrative development examples; summary scores include the complete stated cohort.